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JAVA COLLECTIONS FRAMEWORK

Arrays

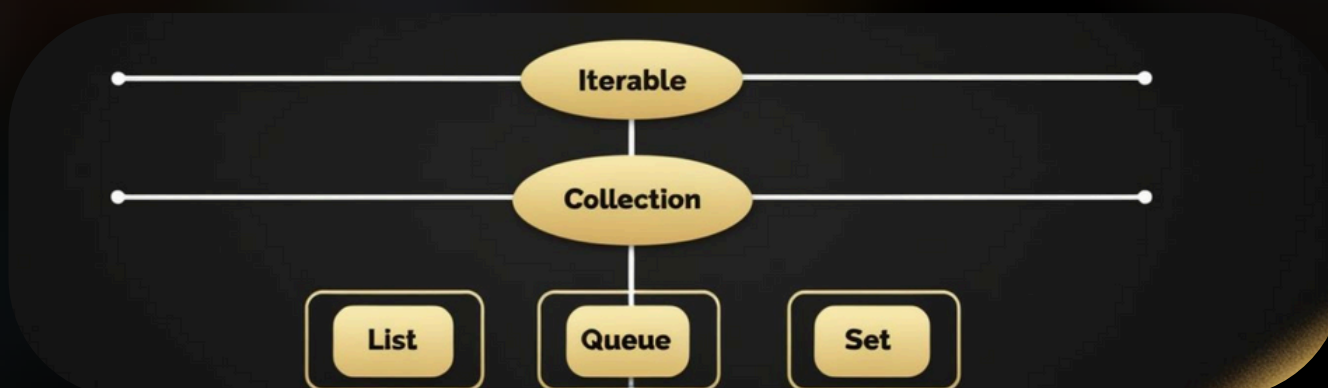
Collection of variables of same type [Limited in size]

Limitations Of array:

1. **Limited size:** we can't store more than specified size of array.
2. **Not Dynamic:** Array is can't grow **dynamically**.
3. **Only same type of elements:** we can't store different type of elements using array.
4. **Data structure:**
 - Using Arrays, we always store elements linearly
 - We can't store information in different data structure
5. **Algorithm:** Don't have inbuilt algorithm support for searching and sorting

When arrays are preferable?

- If we know the exact size of elements and what type of elements, we need to store
- If we want to store the elements linearly, then array is best



About Java Collection Framework

In other programming language like C or C++:

- If we want to store 'different types of data' in 'different data structure' like stack, queue or List we need to create our data structure
- But in Java, programmer no need to create data structure
- By Default, java supports different types of readymade data structure
- This readymade data structure is called "java Collection Framework"
- By using this we can store objects with different data structure
- It is available Since JDK 1.2 in java.util

Benefits of Collections Framework:

- **Reducing the programming effort**
 - readymade data structure
 - No need to create data structure from scratch
 - it reduces programmer efforts
- Reduces effort to learn new data structure
- Increases programming speed and quality
- **Algorithm:** Supports different inbuilt algorithm support for searching and sorting
- **Manipulate objects:** provide different inbuilt methods to manipulate objects
- **Not Limited size:** we can store more than specified size of elements
- **Grows Dynamically:** current size exceeds
- **Types:** Collection can store different type of elements
- **Data Stricture:** we can store information in different data structure using collections
- **Algorithm:** Supports different inbuilt algorithm support for searching and sorting



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JAVA COLLECTIONS FRAMEWORK

What are collections

Definition:

- Collection is an entity that represents group of objects
- In general Collections is a container of objects
- **TYPE:** Objects may be same type or different type
- **Data Structure:** By using this we can store objects with different data structure (List, queue, set, Map)

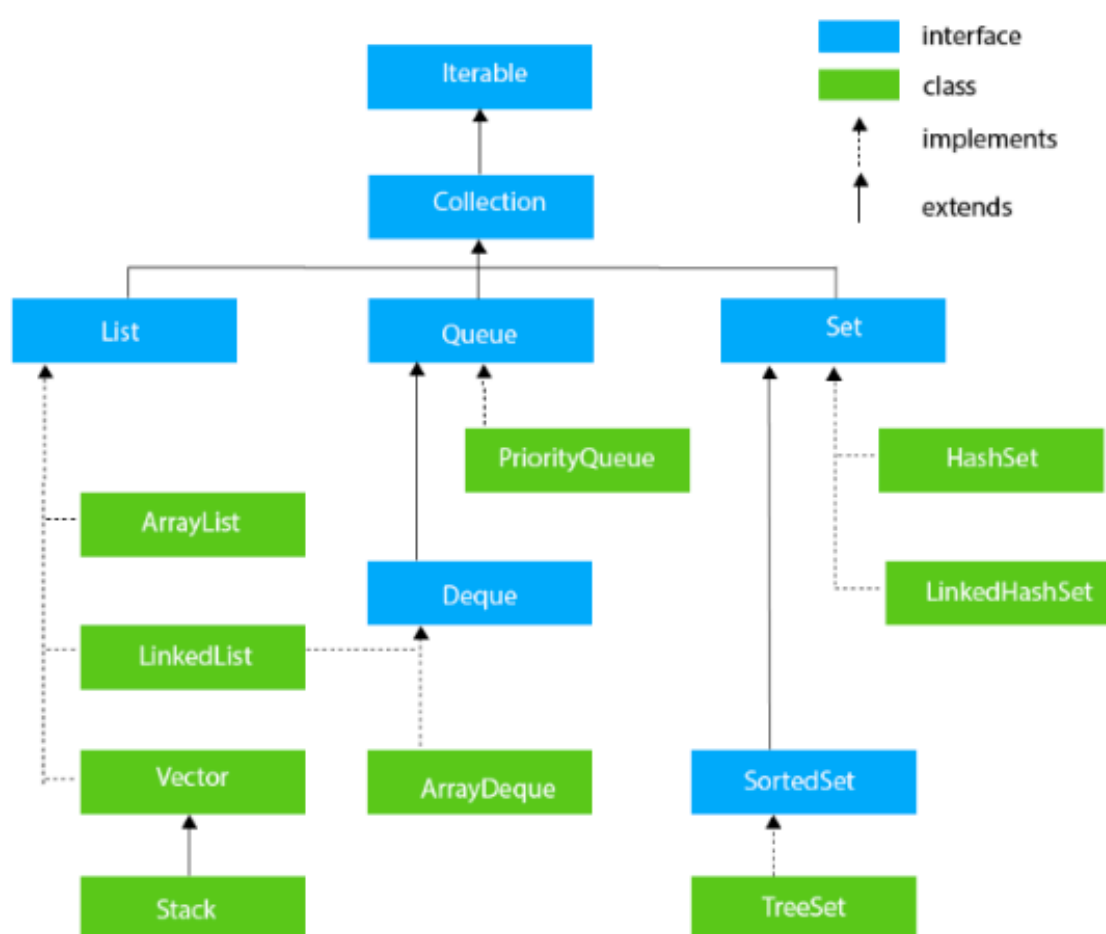
Why Collections over array | Benefits of collections over array | or Benefits of collections

- **Not Limited size:** we can store more than specified size of elements
- **Grows Dynamically:** current size exceeds
- **Types:** Collection can store different type of elements
- **Data Structure:** we can store information in different data structure using collections
- **Algorithm:** Supports different inbuilt algorithm support for searching and sorting

When we use collections over array

- When we not aware of Exact size of elements and the type of elements
- And if we want to store elements in different data structure then we can use collections over array for better performance

Collection Hierarchy



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JAVA COLLECTIONS METHODS

Important methods of collection framework

Methods of Collection interface

public boolean add(Object element):

- Used to add an element into collection.
- If success returns true, otherwise returns false.
- Element can be any type of object.

public boolean addAll(Collection c):

- addAll() method is a bulk operation method of collection interface
- Add all the elements of given collection into current collection
- If success returns true, otherwise returns false
- Collection can be any type of object

public boolean remove(Object element):

- Used to remove an element from the collection.
- If the specified element found in the collection removes and returns true, otherwise returns false.
- Element can be any type of object.

public boolean removeAll(Collection c):

- removeAll() method is a bulk operation method of collection interface.
- It Removes the common elements from the current collections which are also found in the given collection.
- If success returns true, otherwise returns false.

public boolean retainAll(Collection c):

- retainAll() method is a bulk operation method of collection interface.
- It retains the common elements of the current collections which are also found in the given collection.
- If success returns true, otherwise returns false.

public int size():

- Returns the number of elements present in the collection.
- If collection is empty returns 0.

Methods of Collection interface

Public boolean isEmpty():

- It checks whether collection is empty or not.
- If collection is empty returns true, otherwise returns false.

public void clear()

- Removes all elements from the collection and makes collection empty.

public boolean contains(Object element)

- Used to check the specified element in collection present or not.
- If specified element present in the given collection return true, otherwise return false.

public Object[] toArray()

- Used to convert the elements of collection into array of object class type.

public boolean containsAll(Collection c)

- containsAll() method is a bulk operation method of collection interface.
- Used to add all the elements of given collection into current collection.
- If success returns true, otherwise returns false.

public Iterator iterator()

- On invoking this method, it will return the object of implementation class of iterator interface.
- One element at a time.



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LIST COLLECTIONS FEATURES

- List is a type of collection
- **When to use:**
 - If we want to Store the elements based on indexes.
- **Operations on list**
 - All operations on List elements are Based on index only (insert, remove, update, replace)
- **How List works internally**
 - Whenever we add the element into the list
 - List provides an index to the element and stores it with index
- **Duplicate element:** Allowed
- **Null Insertion:** allowed.
- **Preserve the Order of insertion:** preserve the order of insertion until it is disturbing.

Types of Lists

- **Sequential list:** Elements are stored one after another [Array list and vector]
- **Non-Sequential list:** Elements are stored randomly [Linked list]
- But both the list stores the elements on index basis



QUEUE COLLECTIONS FEATURES

Features of Queue collection

- Queue collection stores the elements in FIFO order
- First element is removed first, and last element is removed at last
- Generally, stores the elements without indexes
- We can store with index also with the help of Linked list implementation of queue
- **Duplicate:** Allowed
- **NULL:** Not allowed

Fundamental Operations of Queue

- **Enqueue operation:** Add the element at tail end
- **Dequeue Operation:** Remove the element from front end



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SET COLLECTIONS FEATURES

Set Collection

Features:

- Set collection type of collection stores only unique elements
- It stores the elements without index
- **Insertion order:** Sets doesn't follow insertions order
 - Elements are randomly stored in set collection
- **Duplicate:** Not allowed
- **NULL:** allowed but only one

How Set collections works internally or How set collection does not allow duplicate elements

- Based on the hashCode of an object set allows only unique elements
- Whenever we try to add new element using add () method
 - Add () Method internally calls equals () method
 - equals () method compare the hashCode of new object with hashCode of all the existing set elements
- If equals () method returns true ==> New elements will be ignored
- If equals () method returns false==> New elements will be added into set

How Set collections grows Dynamically:

- Set collections grows dynamically based on load factor OR Fill ratio.
Example:
- The default capacity of HashSet is 16 elements.
- Assume that load factor is 0.75.
- When set elements exceeds current size it will grows dynamically 75% of current capacity
- i.e., it will grow 12 elements



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PURPOSE OF THIS DOCUMENT

- **Use this PDF as a quick last-minute revision guide only.**
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